COURSEWORK 2

LO2: Select appropriate type of renewable energy source and energy storage to mitigate risks associated with security of electrical supply for a given application
LO7: Develop pragmatic knowledge of renewable energy sources and systems.

Overview

You are required to present a ***20-minute oral presentation on off-grid renewable-energy system for a health centre***. This coursework counts towards 25% of the total module grade. The presentation slide is to be submitted by 15:00 on 3 Dec 2026. Late submissions will incur a penalty. The poster should be uploaded as a .ppt, .pptx or .pdf and named as 'FirstName_ID_ CW2'. The presentation schedule will be released at due course.

Please note: *Any generated content using artificial intelligence tools such as ChatGPT is strictly prohibited and will be treated as* ***academic misconduct****.*

Scenario

Many rural and remote communities around the world do not have access to an electrical grid but still require a reliable electricity supply for essential services such as healthcare.

In this case study, you are required to develop an outline design for a renewable-energy-based electrical system to supply an off-grid health centre. The health centre must be supplied entirely by an off-grid electricity system. The electricity demand of the health centre is provided in Figure 1. You may assume that the same daily load profile occurs every day throughout the year.

Continuity and reliability of supply are critical. Your system design must therefore consider intermittency of renewable energy sources and the role of energy storage in maintaining security of supply.

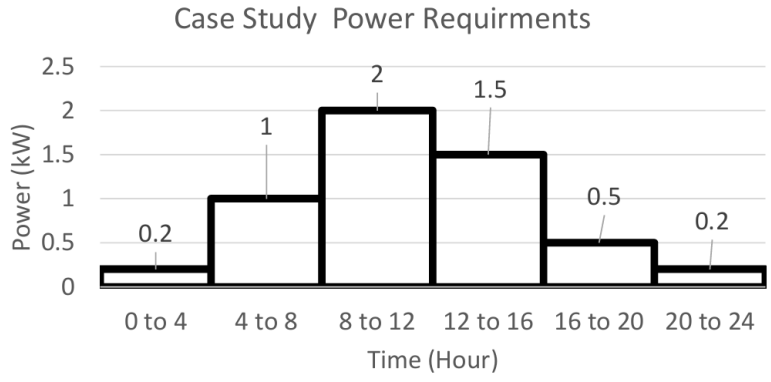


Figure 1: Power required at each four-hour interval.

Location

The coordinates assigned to your health-centre location are provided in the "CW2 Locations" document available on Moodle. The following guidance applies when selecting and interpreting your location:

- The area represented by one degree of latitude or longitude is relatively large. Therefore, there is some flexibility in selecting the precise location of the health centre. You should justify your choice of the specific site.
- If your assigned coordinates fall in the sea, try using the corresponding western longitude and/or southern latitude. If this still produces an unsuitable location, select the nearest coastline or island to the original coordinates.
- If your assigned location is in a populated area or an area that would normally be expected to have access to an electrical grid, ignore the existing grid connection and assume that the health centre must be supplied by an off-grid system.
- If the assigned location is in an unusual or remote environment, such as a mountainous or sparsely populated area, you should still use the assigned location. No justification for the location is required in this case.
- The locations have been randomly generated. If there is a specific reason why you cannot use your assigned location, please contact the module convenor before selecting an alternative location.

Task

1. Energy Supplies

Investigate the climate and renewable-energy resources available at your assigned location and provide a high-level assessment of their potential. Your assessment should consider all renewable energy sources covered in the module, including sources that may have limited or negligible potential at your chosen location.

You should provide evidence to demonstrate why a particular resource is or is not suitable. For example, if hydropower is considered unsuitable because of limited rainfall or water availability, provide relevant data to demonstrate this rather than simply excluding hydropower from the assessment.

Use figures, graphs, maps and tables where appropriate to present the available resource data. Examples include:

- annual solar irradiance;
- monthly precipitation or rainfall;
- wind-resource data;
- geothermal-resource maps; and
- other relevant resource indicators.

Based on your assessment, select the most appropriate renewable energy source, or combination of sources, for supplying the health centre. Clearly identify the preferred source(s) and justify your decision using both qualitative and quantitative evidence. You must also select a second renewable energy source and use it as a basis for comparison with your preferred option.

Resource data should be obtained from reliable meteorological, climate and renewable-energy databases, websites and atlases. All sources must be appropriately referenced.

Use of Generative AI:
Generative AI may be used to assist in finding and refining ideas in this section. However, the assessment is expected to be supported by appropriate figures, maps and charts based on actual resource data. These should be produced using appropriate data sources and analysis tools rather than generated by AI.

2. Energy Conversion
Develop an outline design of the energy conversion and storage system required to supply the health centre. Your design should specify the major components of the system, which may include:

- renewable-energy generators;
- power-electronic converters and/or inverters;
- energy-storage systems;
- generators or other backup supplies, where required; and
- other relevant system components.

The proposed system should be presented using a system diagram developed specifically for your design. The diagram should clearly show the energy flow and the main components of the proposed system. Include the relevant conversion efficiencies and calculate the approximate electrical power and/or energy available at the appropriate stages of the system.

The renewable-energy generation and energy-conversion equipment must be appropriately sized to meet the electrical requirements of the health centre. The required energy-storage capacity must also be calculated and justified based on the energy demand and availability of the renewable resource.

Provide the basic technical specifications of the major equipment used in your proposed system.

Use of Generative AI:
Generative AI may be used in this section. However, this section involves significant mathematical calculations, and all AI-generated calculations must be checked and verified independently. The system diagram must be produced by the student and should not be an AI-generated diagram.

3. Costs, CO_2 Emissions and Comparisons

Use a Generative AI tool to estimate the cost of the proposed renewable-energy system for your assigned location. Each cost estimate should be supported by an appropriate reference or source where possible. The costing should include:

- all major system components;
- the estimated cost of each component;
- initial/start-up costs;
- operating and running costs; and
- expected system/component lifetime.

Next, use the Generative AI tool to estimate the CO_2 emissions associated with manufacturing and operating the proposed renewable-energy system based on the system size calculated in Section 2. Compare these emissions with those associated with generating an equivalent amount of electrical energy using a fossil-fuel-based system.

Finally, use the AI tool to compare your proposed renewable-energy system with another energy-generation technology, which may be a non-renewable technology. You must make more than four comparisons between the two systems. Each comparison should include:

- a numerical comparison supported by appropriate evidence; and
- a clear statement explaining which system performs better for that particular criterion.

For each of the AI-generated analyses, you must critically evaluate the results produced by the AI tool. Your critique must be supported by appropriate references and evidence. Identify:

- aspects of the AI-generated results that you consider to be reliable or correct;
- aspects that may be incorrect, uncertain or potentially misleading;
- assumptions made by the AI tool that may affect the results; and
- how any identified errors or limitations could be corrected.

If you choose not to use a Generative AI tool, you may perform the same analyses independently. You will still be eligible for the full range of marks; however, the Generative AI critique component will not apply.

Presentation Slide Requirements

Content: Visual and concise presentation
Duration: 15 minutes (presentation), 5 minutes (Q&A)